

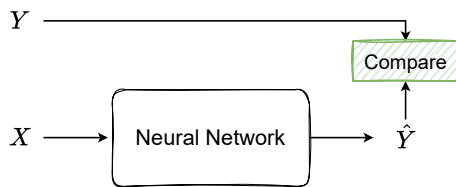
DEEP LEARNING

Multimodal Deep Learning

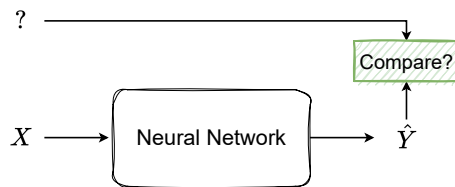
Subhankar Roy

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University of Bergamo

Recap: Learning Paradigms



Supervised Learning



Unsupervised Learning

- **Supervised learning** is expensive as it requires collecting labelled training data (X, Y) .
- A more cost-effective way is to collect a lot of images and learn from these images alone, ie, $(X, ?)$. This is known as **unsupervised learning**.

Recap: Representation Learning

- Under the umbrella of unsupervised learning, representation learning (or **self-supervised learning**) involves **automatically discovering** useful features or **representations** from raw data (eg, images alone) to improve performance on tasks like classification.
- We develop a **pretext task**, or a "fake" objective, that allows us to learn useful features, without requiring class labels. They are:
 - ▶ Rotation prediction
 - ▶ Predict relative patch locations
 - ▶ Solving jigsaw puzzles
 - ▶ Predict missing pixels (or inpainting)
 - ▶ Image coloring, video coloring
 - ▶ A more general pretext task that is based on the idea of **contrasting** *positive pairs* from *negative pairs*

Recap: Contrastive Representation Learning



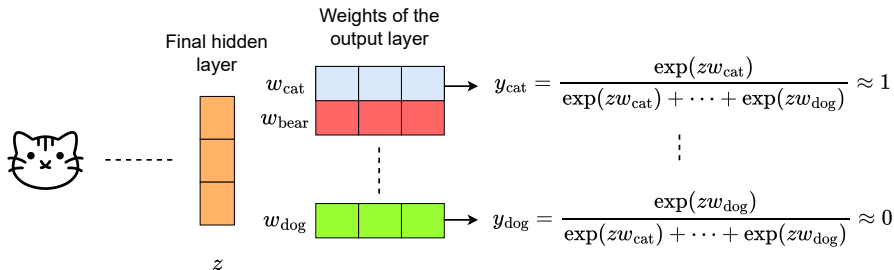
- Formally, we want:

$$\text{score}(f(x), f(x^+)) \gg \text{score}(f(x), f(x^-)),$$

where x : reference sample, x^+ : positive sample, and x^- : negative sample.

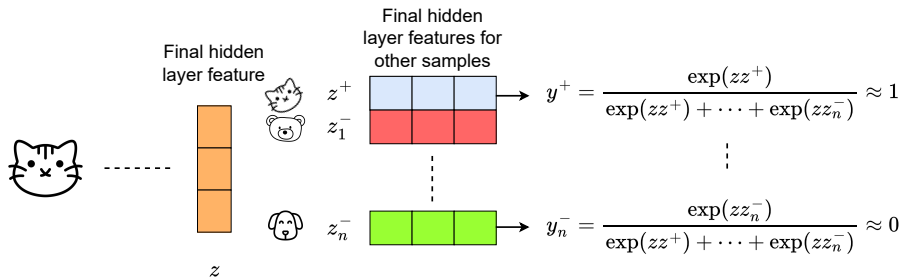
- Given a chosen score function, we aim to learn an encoder function f , ie, a neural network, that yields high score for positive pairs (x, x^+) and low scores for negative pairs (x, x^-) .

Recap: MLP Classifier



- In the output layer of a MLP, we:
 - ▶ Compute the dot product between the final hidden layer feature representation z and the weights corresponding to the class cat: w_{cat} .
 - ▶ Similarly, we compute the dot products between z and the weights of all other classes.
 - ▶ For a given output neuron, the output, ie, y_{cat} , is computed as the ratio of zw_{cat} and the sum of its dot product with all other classes (with an $\exp$).

Recap: Contrastive Representation Learning

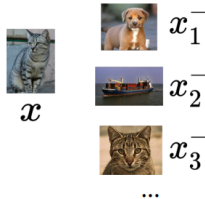
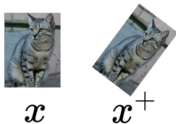


- In contrastive learning, the difference is that the weights of the output layer are replaced by the features of other samples in the mini-batch:
 - ▶ The positive sample z^+ of the reference sample z
 - ▶ The negative samples $\{z_1^-, \dots, z_n^-\}$, coming from all other n samples in the mini-batch
- Therefore, the output layer of contrastive representation learning can be viewed as a classifier, without actual weights.

Recap: Contrastive Representation Learning

Loss function given 1 positive sample and $N - 1$ negative samples:

$$\mathcal{L} = -\log \frac{\overbrace{\exp(s(f(x), f(x^+)))}^{\text{score for the positive pair}}}{\underbrace{\exp(s(f(x), f(x^+)))}_{\text{score for the positive pair}} + \underbrace{\sum_{j=1}^{N-1} \exp(s(f(x), f(x_j^-)))}_{\text{score for negative pairs}}}$$



Recap: Augmentations for Generating Positives in SimCLR¹



(a) Original



(b) Crop and resize



(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate $\{90^\circ, 180^\circ, 270^\circ\}$



(g) Cutout



(h) Gaussian noise



(i) Gaussian blur

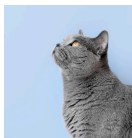


(j) Sobel filtering

¹SimCLR paper

Today

- A new paradigm of representation learning, where we will train on data that come from two **different modalities**: image and its corresponding caption.
- Such data is cheap to collect. Go to the internet, scrape images, and most images are associated with captions, tags or alt-text.



Image

Caption

```
src-detail-preview' style="--detail-preview-max-height: calc(100vh - 285px); --detail-preview-max-height-mobile: calc(100vh - 225px); aspect-ratio: 1 / 1.48045;"/>  

```

Inspect element (in browser)

- It has become the de facto recipe for training latest and greatest models.
- Vision Language Models (VLMs) we will cover:
 - ▶ CLIP
 - ▶ BLIP
 - ▶ LLaVA

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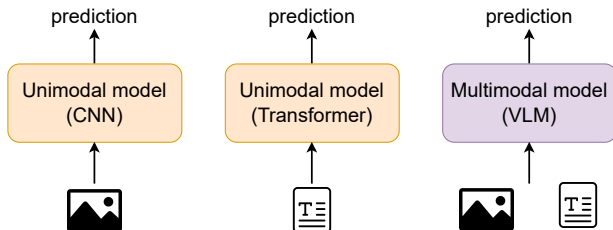
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Vision Language Models

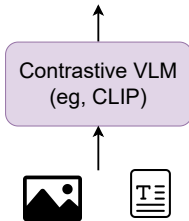
- So far, all the models that we have seen can be regarded as **unimodal models** because these models take as input only one modality of input data:
 - ▶ CNNs can take images as input
 - ▶ Transformers can take text/images as input
- We have another family of models that can process more than one modality of input data *at the same time*: text, images, videos, etc. These models are called **multimodal models** or **vision-language models (VLMs)**.



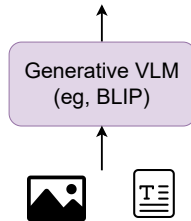
Vision Language Models

- There are a wide variety of VLMs in the literature. They can be roughly categorized into two types:
 - ▶ **Contrastive VLMs**. These VLMs are trained solely using contrastive loss and can be used for tasks such as classification and retrieval. Very similar to a regular MLP/CNN.
 - ▶ **Generative VLMs**. These VLMs are trained with a combination of contrastive (optional) and language modelling (LM) loss. Due to the use of LM loss, it can generate sequences (eg, text, but not limited to text).

prediction
(classification, retrieval)

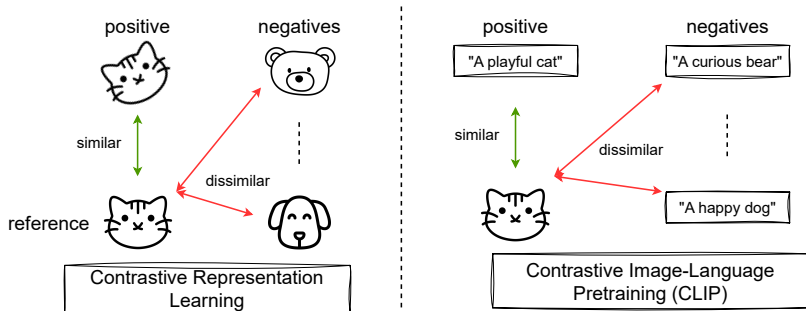


prediction
(generation)



Contrastive Vision-Language Models

Contrastive Language-Image Pretraining (CLIP)



- Left: Contrast between the representation of an image with positive and negatives, both derived from images.
- Right: Contrast between the representation of an image with positive and negatives, where the positives and negatives are derived from **captions** (or text).

CLIP: The Benefits of Using Language

- **Question:** What is the advantage of incorporating language in contrastive learning, when we could formulate contrastive loss using images alone?
- **Answers:**
 - ▶ Language is a richer form of supervision, and leads to learning better representation²
 - ▶ We can use natural language to create classifiers of any kind (also called as **zero-shot classification**). More on this later.

labels = ["Smile", "Sad", "Cry", "Surprise", "Anger"]

Predicted Class: Surprise



labels = ["Bald Eagle", "Golden Eagle", "Vulture", "Hawk"]

Predicted Class: Bald Eagle

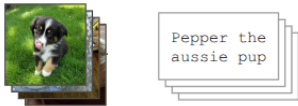


- ▶ Enables cross-modal retrieval. More on this later.

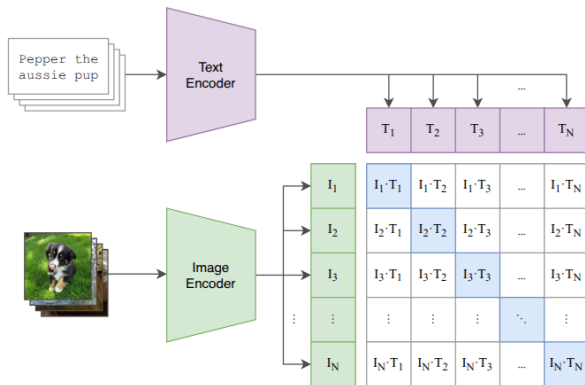
²CLIP paper

CLIP: Training Data

- To collect training data, we scrape images and their captions from the internet, followed by some filtering.
- In CLIP, the training data is of the format: $(\mathbf{x}, c)$, where $\mathbf{x}$ is the image and c is the corresponding caption.
- Quite similar to the training data in supervised learning, where we had $(\mathbf{x}, y)$, where y is the class label.
- Sometimes, we call this paradigm **webly-supervised learning** since we are using data from the web to train our models.
- During training, we use mini-batches of $(\mathbf{x}, c)$ pairs.

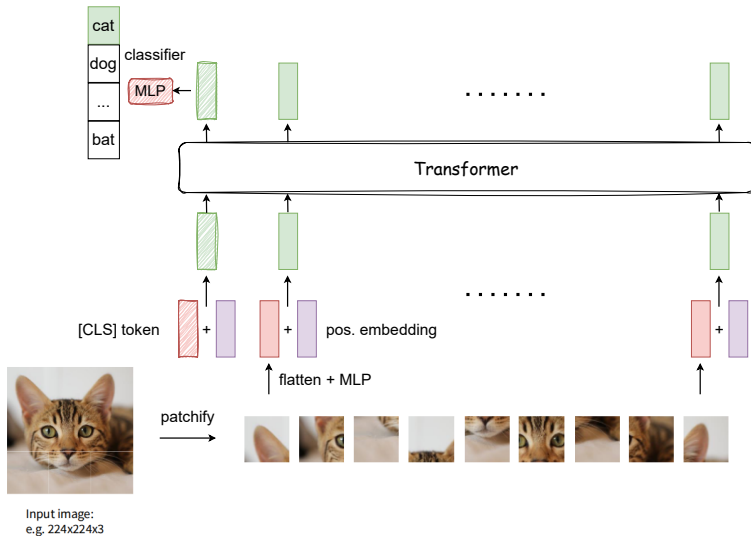


CLIP: Overview

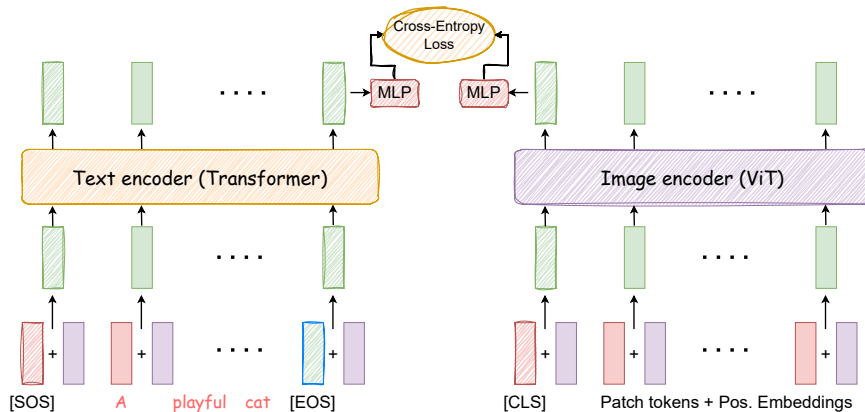


- CLIP maximizes the similarity between the representation of an image I_1 and its corresponding caption T_1 , and minimizes the similarity between I_1 and all other captions in the mini-batch.

Recap: Vision Transformer (ViT)

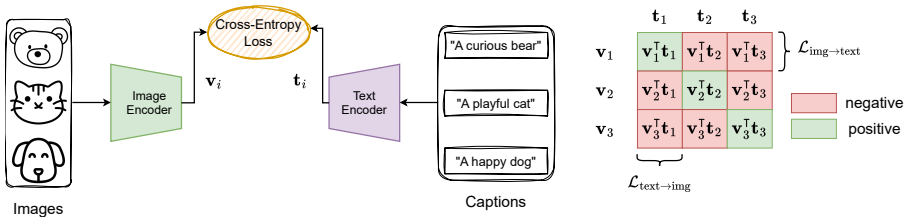


CLIP: Architecture



- During training, we align the representations from the [EOS] token of the text encoder (a transformer) and the [CLS] token of the image encoder (a ViT) for a given (x, c) training pair.

CLIP: Training

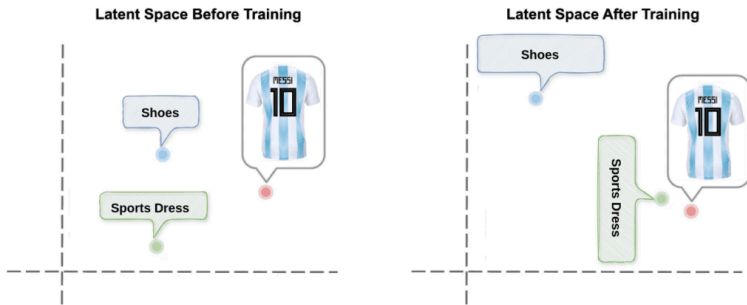


- Let the image representation be $\mathbf{v}_i$ and the text representation be $\mathbf{t}_i$ for the i -th training point. CLIP's objective function is the sum of the two objective functions:

$$\mathcal{L}_{\text{img} \rightarrow \text{text}} = -\frac{1}{3} \sum_{i=1}^3 \log \frac{\exp(\mathbf{v}_i^\top \mathbf{t}_i)}{\sum_{j=1}^3 \exp(\mathbf{v}_i^\top \mathbf{t}_j)} \quad \text{image-to-text loss}$$

$$\mathcal{L}_{\text{text} \rightarrow \text{img}} = -\frac{1}{3} \sum_{i=1}^N \log \frac{\exp(\mathbf{v}_i^\top \mathbf{t}_i)}{\sum_{j=1}^3 \exp(\mathbf{v}_j^\top \mathbf{t}_i)} \quad \text{text-to-image loss}$$

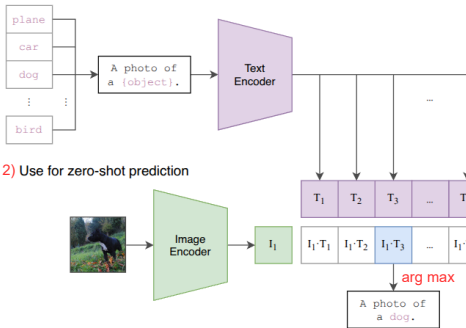
CLIP: Latent Image-Text Representation Space



- Through training, we are trying to enforce:
 - ▶ the representation of an image to be closer to the representation of its caption.
 - ▶ the representation of image to be far from the representation of unrelated captions.
- In the above example, the image of the jersey is closer to the caption "Sports Dress" than to the caption "Shoes" in the latent representation space.

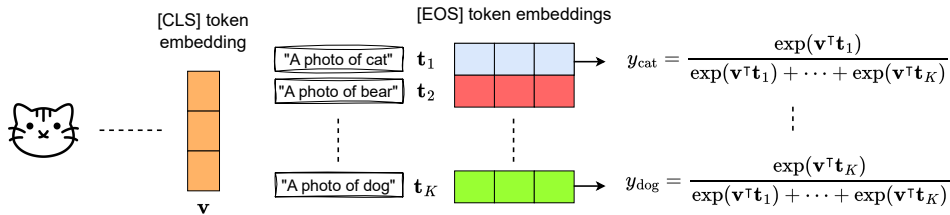
CLIP: Inference

(Step 1) Create dataset classifier from label text



- Step 1: Given a test image, construct the text embeddings from the class names, using a pre-defined template "a photo of a [CLASS NAME]". All the text embeddings together forms the classifier.
- Step 2: Compute the similarity between the image embedding and all the text embeddings, and the prediction is the class that has the highest similarity.

Analogy between CLIP Inference with MLP Classifier



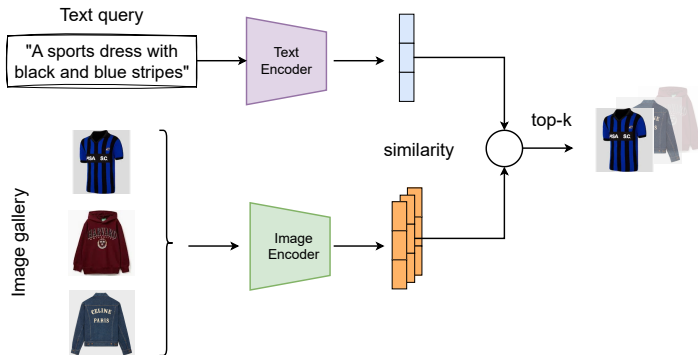
- Recall, in a MLP classifier, we compute the dot product between the image representation and the weights of all the classes.
- In CLIP, we have a similar inference mechanism, except the weights are replaced by the text embeddings corresponding to the text (also called template) "a photo of a [CLASS NAME]". The text embedding corresponds to [EOS] token.
- We are not limited by the above caption, we can use any kind of template we want.

CLIP: Zero-Shot Classification

	Dataset Examples						ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet							76.2	76.2	0%
ImageNetV2							64.3	70.1	+5.8%
ImageNet-R							37.7	88.9	+51.2%
ObjectNet							32.6	72.3	+39.7%
ImageNet Sketch							25.2	60.2	+35.0%
ImageNet-A							2.7	77.1	+74.4%

- Due to its flexibility in the classifier and large scale pre-training (trained on 400m images), CLIP can classify (almost) any dataset → **zero-shot classification**.
- Above figure shows that CLIP can outperform models (eg, ResNet101) that have been trained on labelled samples of ImageNet dataset.

CLIP: Image/Text Retrieval



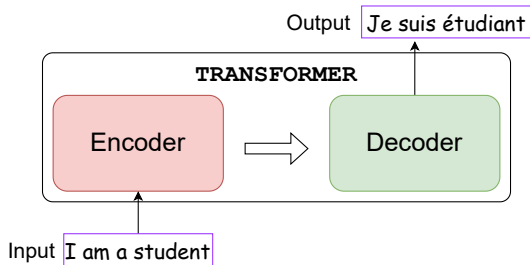
- We can use CLIP for retrieving images from an image gallery based on a text query, ie, searching images based on text, called **text-to-image retrieval**.
- We can do the reverse, by retrieving text based on a query image, which is called **image-to-text retrieval**.

Generative Vision-Language Models

Generative Vision-Language Models

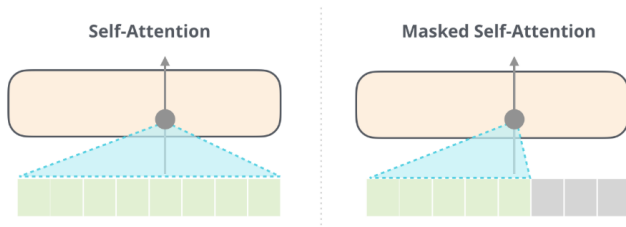
- CLIP are vision-language models, where we are asking the question: "Are these a match?"
- Thus, CLIP can only be used for tasks such as classification or retrieval.
- What if we wanted to ask for more, questions like: "What is in the image?" CLIP can not handle such kind of tasks.
- Therefore, we will move from simply "matching images/text" to "producing text from images"!
- These are made possible by [generative vision-language models](#), such as BLIP, LLaVA.
- It has many applications:
 - ▶ Image captioning
 - ▶ Visual question answering
 - ▶ Holding a dialogue (eg, a personal assistant)

Recap: Encoder-Decoder Transformer



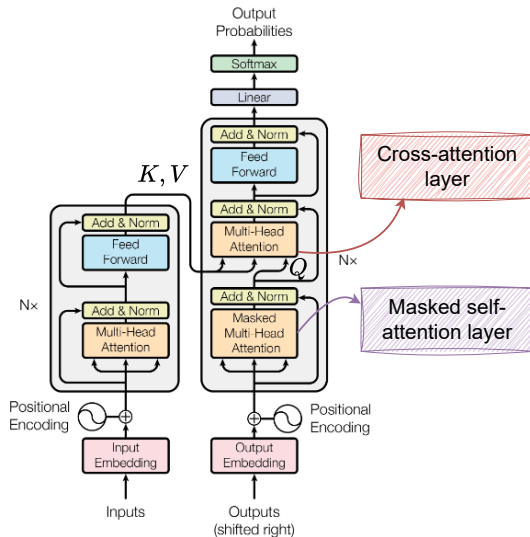
- The original Transformer has two main building blocks: Encoder and Decoder
- There are three main kinds of components inside these blocks:
 - ▶ **Self-attention** (in Encoder): which can attend to all the tokens in a given sentence. Q, K, V comes from the same sequence.
 - ▶ **Causal self-attention** (in Decoder): which can attend to all the tokens up until its current location. Q, K, V comes from the same sequence.
 - ▶ **Cross-attention** (in Decoder): similar to SA, except the Q comes from the decoder, and K, V comes from the encoder.

Recap: Self-Attention versus Masked Self-Attention



- **Masked self-attention** (or causal self-attention) is a mechanism that restricts which tokens a model can attend to during training or inference.
- Let's say the model has to predict the next word "buy" for this input sentence: "I went to the store to buy some milk" (Gray means not seen so far).
- If self-attention can see the entire sentence, then it can cheat by looking at "milk".
- Masked self-attention hides away the future segment (the one in gray). Now the model must infer "buy" based only on prior context.

Recap: Cross-Attention



Recap: Cross-Attention

- In the case of language translation task (eg, en $\rightarrow$ fr), the model still needs a way to connect back to the input sentence (in en) the encoder processed.
- Cross-attention allows the decoder to **look over** at the encoder's output to decide which parts of the input matter the most for generating the next word.

target sequence: il a balancé la batte avec une force incroyable <EOS>

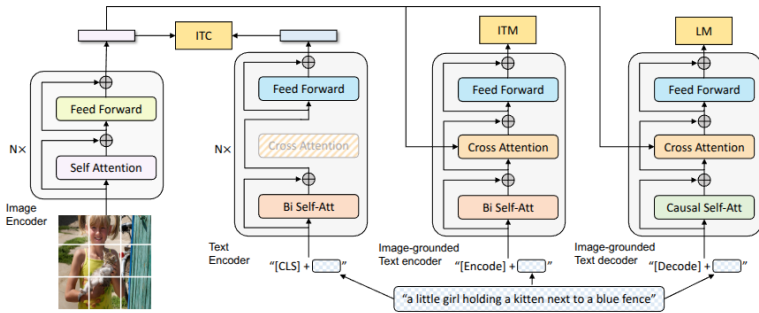
source sequence: he swung the bat with incredible force <EOS>

input to the encoder: he swung the bat with incredible force <EOS>

input to the decoder: <start> il a balancé la batte avec une force incroyable <EOS>

- The encoder's output provides the key (K) and value (V) to the decoder's cross attention layer.
- The query (Q) comes from the decoder's own hidden states (based on what's been generated so far).

BLIP: Bootstrapping Language-Image Pre-training



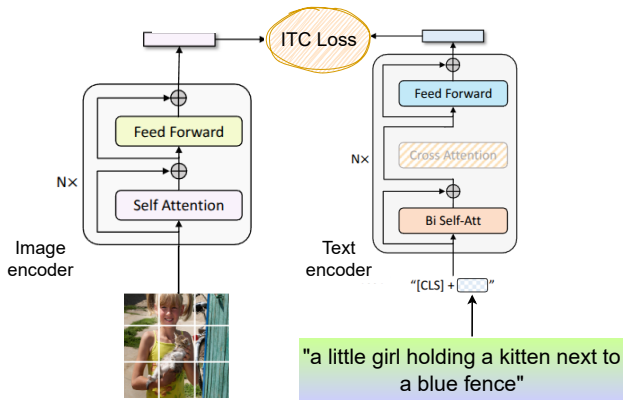
- BLIP, which stands for Bootstrapping Language-Image Pre-training, is in part similar to CLIP, except the model is also trained to **predict the caption**, besides matching an image with its caption.
- Because of this pre-training, BLIP can also generate captions given an image, which CLIP can not. Therefore, BLIP is a generative model.

BLIP: Training Objectives

- BLIP³ is trained with three objective functions:
 - ▶ **Image-text contrastive** (ITC) loss: this aligns the vision and language representation. Similar to the loss used in CLIP.
 - ▶ **Image-text matching** (ITM) loss: similar to the CLIP loss, except it models vision-language interactions with the cross-attention layers. Essentially, it distinguishes between positive and negative image-text pairs.
 - ▶ **Language modelling** (LM) loss: the network is trained to generate captions given images. Leverages the causal self-attention (or masked self-attention).

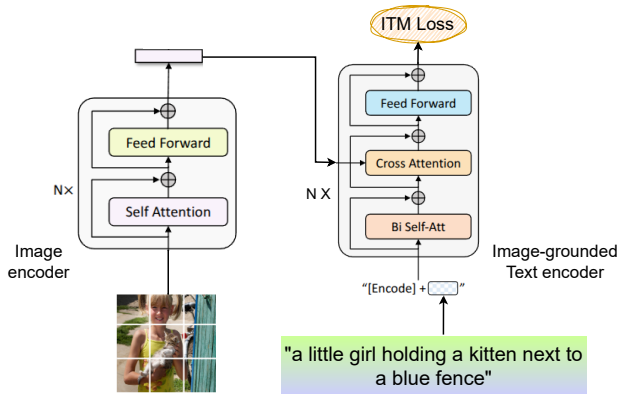
³BLIP paper

BLIP: Image-text Contrastive (ITC) Loss



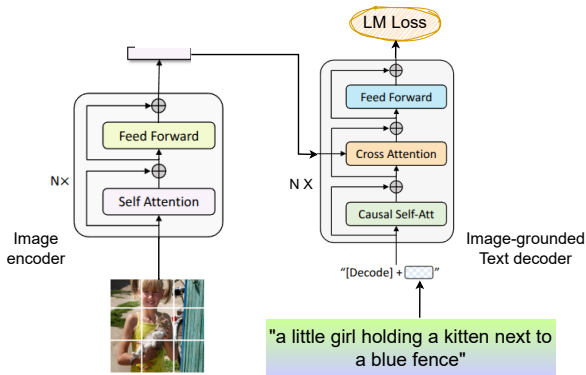
- ITC loss encourages the representation of the image to be similar (or aligned) with the representation of the text (or the caption).
- This is similar to the similarity loss that is used for training CLIP.

BLIP: Image-text Matching (ITM) Loss



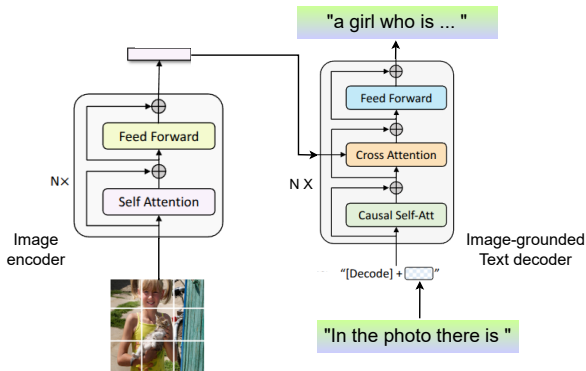
- Image-grounded text encoder inserts one additional cross-attention layer that models vision-language interactions.
- ITM loss is a binary classification loss for distinguishing whether an image-text pair is positive (matched) or negative (unmatched).

BLIP: Language Modelling (LM) Loss



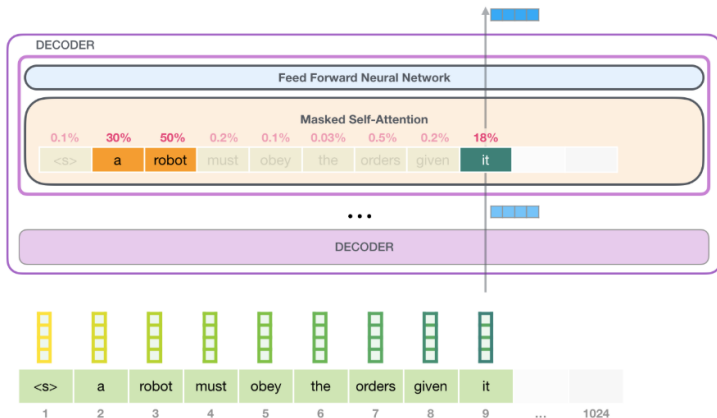
- The LM loss aims to generate textual descriptions given an image, as in autoregressive language models.
- Note that we replace the self-attention layers with causal self-attention layers because we do not want the decoder to attend to future tokens in the caption while generating captions.

BLIP: Inference



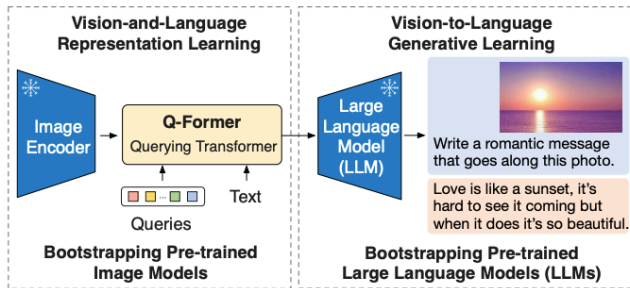
- During inference, if we want to generate something new, we can:
 - ▶ Either simply provide the [DECODE] (or [SOS]) token and the model will autocomplete the caption.
 - ▶ Or we can provide the starting few tokens, and the model will autocomplete it.
- The decoder outputs tokens until the [EOS] token is reached.

Recap: Decoder-only Transformer



- GPT, a decoder-only transformer, predicts the next word in a sequence using large amounts of text, learning language patterns through unsupervised pre-training.

BLIP2



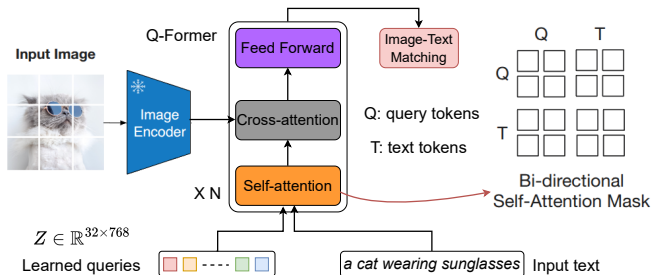
- BLIP2⁴ is a follow-up work of BLIP that introduces two novelties in the framework:
 - ▶ Unlike BLIP, does not train the image encoder and large language model (LLM) from scratch. Instead uses pre-trained models which are easy to find.
 - ▶ Introduces Querying Transformer (or Q-Former), a lightweight framework for connecting the image encoder with the LLM.

⁴BLIP2 paper

BLIP2: Pipeline

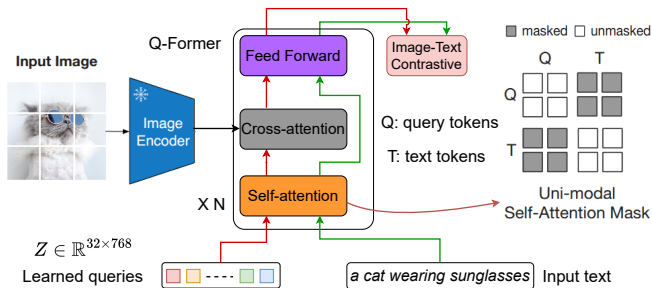
- Q-Former follows a transformer architecture, and it is initialized from BERT model.
- It has two new components than BERT architecture:
 - ▶ Learned queries, which are embeddings $Z \in \mathbb{R}^{32 \times 768}$
 - ▶ cross-attention (CA) layer
- BLIP2 is trained in two stages:
 - ▶ **Stage 1:** In this stage the Q-Former is trained to extract visual representation most relevant to the text. This is done by optimizing the three losses we have seen before in BLIP: ITC, ITM and LM losses. Special attention masks are designed to implement different loss functions.
 - ▶ **Stage 2:** In this stage the Q-Former is connected to a frozen LLM for generating free-form text at the output.

BLIP2: Image-text Matching (ITM) Loss



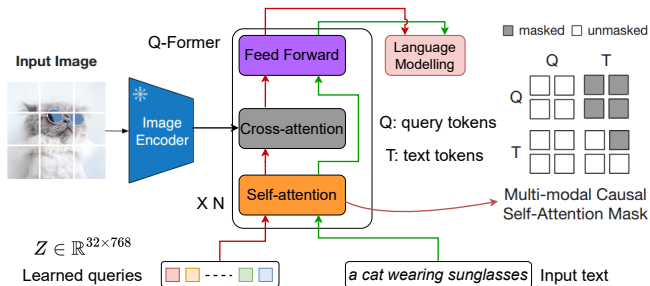
- In the CA layer, the queries interact with the image features ($\mathbb{R}^{257 \times 1024}$) coming from the Image Encoder. The goal is to create a **compressed representation**, ie $\mathbb{R}^{32 \times 768}$, of the image using a fewer learnable queries.
- In the SA layer, the queries interact with each other and with the text tokens. This is implemented by using a **bi-directional self-attention mask** that allows every token and query to attend to each other.

BLIP2: Image-Text Contrastive (ITC) Loss



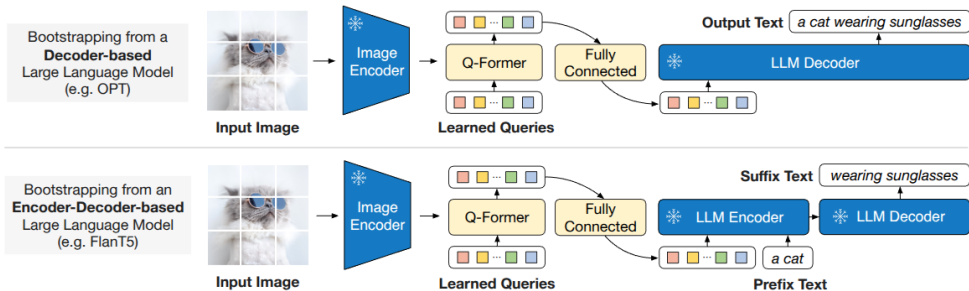
- As the color of the arrows show, we compute two different representations independent of each other for computing the ITC loss:
 - ▶ The image representation (obtained through CA) with the learned queries
 - ▶ The text representation
- A **uni-modal self-attention mask** ensures that the queries can *only* interact with other queries, and the text tokens can *only* attend to other text tokens.

BLIP2: Language Modelling (LM) Loss



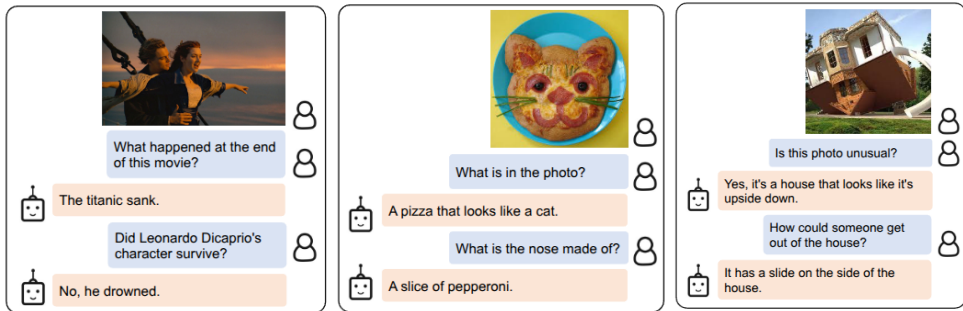
- Similar to the ITC loss, in the language modelling (LM) loss, we compute two independent representations, as indicated by the colors.
- A **multi-modal causal self-attention mask** is designed with the following characteristics:
 - ▶ The queries can *only* attend to the queries.
 - ▶ The text tokens can attend to *both* the queries and text tokens, but *only* the text tokens that are on the left, ie, no future tokens.

BLIP2: Bootstrapping a LLM



- In the second stage, the Q-Former is followed by a fully-connected network, to project the learned queries, which are input to a LLM.
- There are two ways to implement it:
 - ▶ The learnable queries are used as a prefix in a decoder-only LLM (top)
 - ▶ The learnable queries and a text prefix are together used as a prefix in an encoder-decoder LLM (bottom)

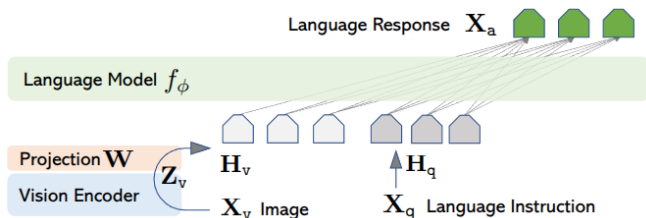
BLIP2: Example Generations



- BLIP2 can be used to hold a conversation given an input image.
- One can ask different kinds of questions: (i) knowledge based, (ii) visual based, and (iii) common sense reasoning.

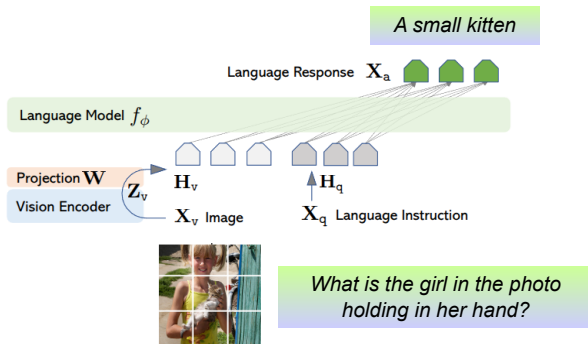
LLaVA

- LLaVA⁵ (Large Language and Vision Assistant) is a generative vision-language model that connects a pre-trained image/vision encoder with a pre-trained large language model. The goal is to train a single model that can generate captions or answer questions given an image.
- LLaVA has the following components: (i) Vision Encoder (frozen), (ii) Projection layer (trained), and (iii) Language Model (fine-tuned).



⁵LLaVA paper

LLaVA: Overview

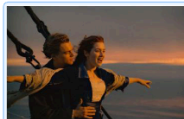


- To train, we need data in the following format: (i) an image, (ii) a question (language instruction), and (iii) an answer (or language response).
- LLaVA is trained using the language modelling loss, like in BLIP and BLIP2. The only difference is that the model is asked to output only the answer, and not autocomplete the input text.

LLaVA: Example Generations



In the image, a young woman is leaning on a man in front of a beautiful sunset, with both of them standing on a boat. They appear to be enjoying a romantic moment together. The scene suggests that they could be depicted as the characters Jack and Rose from the popular movie Titanic. In the movie, their romance blossoms aboard the fictional ocean liner Titanic, with the pair ultimately sharing a passionate kiss as the ship sinks into the ocean. The ending of the movie is known for its emotional and heartfelt scenes, where the characters are forced to say goodbye to each other as they are separated by the sinking ship. The image evokes a sense of romance and nostalgia, paying tribute to the iconic love story portrayed in the movie Titanic.



User

What's the ending of this movie?

Start a new conversation, and the history is cleared.



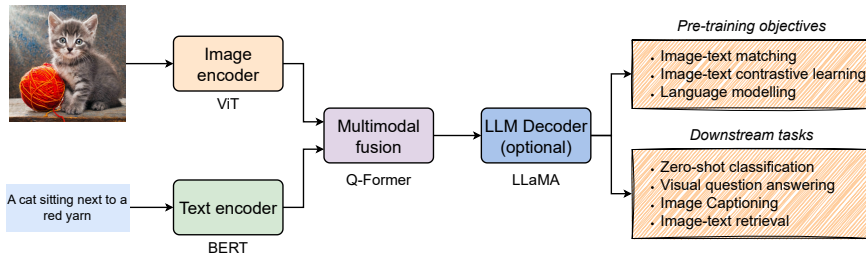
The painting depicts a dog in a humorous situation, where it appears to be wearing a woman's dress and standing in front of a landscape. Although the artist of the painting is not explicitly identified, it could be a reference to the famous artist Da Vinci, known for his artistic works, including the Mona Lisa. The painting is likely a creative and comical take on the traditional portrait style, replacing a human figure with a dog dressed in a woman's clothing, making it an interesting and amusing visual piece.



User

Do you know who drew this painting?

Closing Summary



- We explored a new learning paradigm called multimodal deep learning, which involves learning from two modalities: images and text. Such models are commonly called vision-language models.
- We saw two kinds of VLMs: CLIP-family and BLIP-family. The first can be used for tasks like classification, whereas the second is especially used for generating text from images.

Closing Summary

- Vision-Language Models (VLMs) have become very popular in the field of deep learning because:
 - ▶ Training data, although a little noisy, is easy to collect (freely) from the internet
 - ▶ Leads to stronger representation than what could be achieved with one modality alone
 - ▶ Scales well, ie, more data + bigger models → better models
 - ▶ Results in very useful models that has applications in a wide range of real-world applications
- The area is trendy and ever-growing. See the following for further reading:
 - ▶ A survey⁶ on VLMs
 - ▶ A blog⁷ on VLMs
 - ▶ A tutorial⁸ (Unit 4) on VLMs

⁶VLM survey

⁷VLM blog

⁸Huggingface tutorial